

Simulating the JV curve of a solar cell

Physics and Characterization of Emerging Solar Cells, UZH

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1 Drift-diffusion simulator

In this exercise we will use SIMsalabim¹, an open-source drift-diffusion simulator for solar cells. We run it through its web interface at <http://simsalabim-online.com/>, so no installation is required. We will use the steady-state JV simulator (Fig. 1):

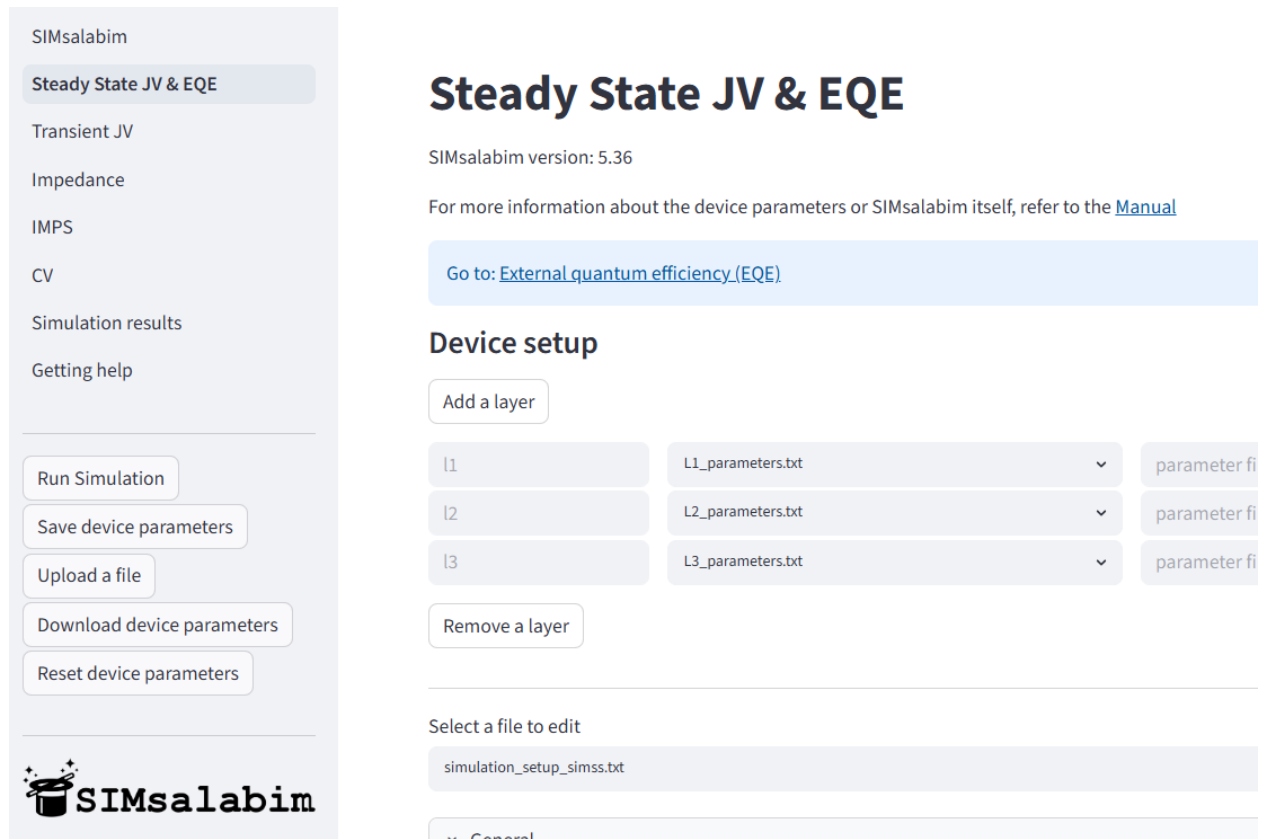


Figure 1: Online user interface of SIMsalabim steady-state JV curve simulator.

2 Device model

We model the solar cell as three semiconductor layers (Fig. 2): the absorber in the middle, where light is absorbed and electron-hole pairs are generated, sandwiched between two charge-selective transport layers, an electron transport layer (ETL) that lets electrons pass while blocking holes, and a hole transport layer

¹<https://github.com/kostergroup/SIMsalabim>

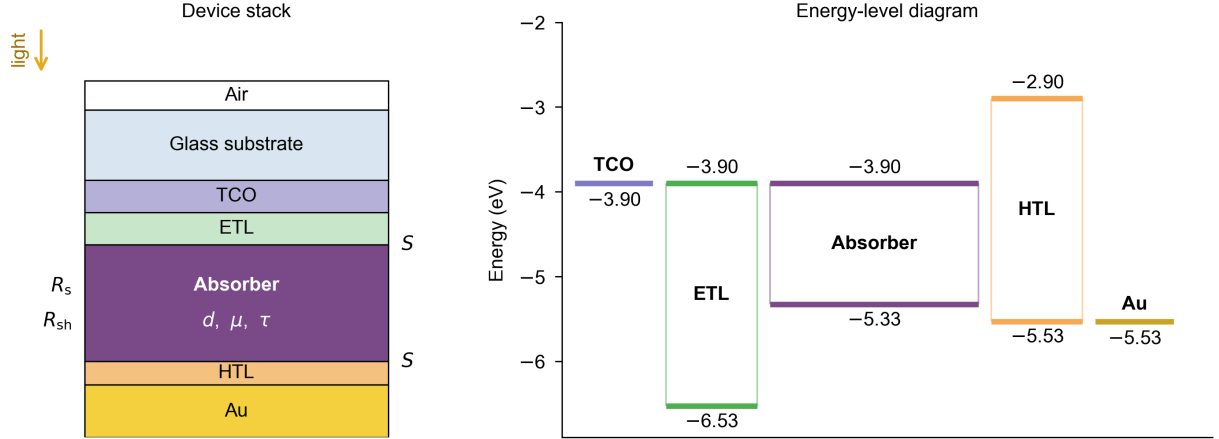


Figure 2: Model device stack and energy-level diagram.

(HTL) that does the opposite. For these simulations, only the absorber layer parameters will need to be modified.

A simulation in SIMsalabim needs the simulation setup file and one parameter file per layer. For the absorber, we also will need to set the optical **nk** file. The rest of **nk** files for layers as well as the incident solar spectrum are already available in the online tool, so they do not need to be uploaded again.

Download the starting input files. From the GitHub repository², download the `bad_start/` folder. You should have:

- `simulation_setup.txt`
- `ETL_parameters.txt`, `ABS_parameters.txt`, `HTL_parameters.txt`
- the `nk_*.txt` files in `Data_nk/` (only needed for absorber)
- the spectrum `AM15G.txt` in `Data_spectrum/` (not needed)

Upload them to the GUI. In the right-hand menu (Fig. 3, left), click **Upload a file**. A dialog opens (Fig. 3, right): select **Simulation setup** from the dropdown, drop in `simulation_setup.txt`, then upload the three layer parameter files (`ETL/ABS/HTL_parameters.txt`) under **Select all the layer parameter files...** The dialog lists any Missing layer parameter files until all three are in. Then click **Submit**. After doing this click on **Save device parameters** to update the model. Repeat the **Upload a file** step with the dropdown set to **nk file** to upload the absorber file `Data_nk/nk_absorber_Eg1.43eV.txt` and again **Save device parameters**.

3 Running the simulations

Run the simulation. Click on **Run Simulation** (top of the side menu, Fig. 3, left). When the solver finishes, go to **Simulation results** in the main menu (Fig. 1, left). The GUI should display the simulated *JV* curve and the key figures of merit: J_{sc} , V_{oc} , FF, MPP.

The obtained *JV* should match the bad start in Fig. 5. The parameters for the absorber in `bad_start/`, listed in Table 1, are not optimum for this solar cell.

²<https://github.com/nsdt-zhaw/jv-dd-simulations>



Figure 3: Uploading inputs. Left: side menu with the Upload a file button. Right: upload dialog — pick the file type from the dropdown, then drag the files in. The Missing layer parameter files list disappears once all three layer files are uploaded.

Change parameters to improve the solar cell. Open `ABS_parameters.txt` in the GUI’s text editor (you can also edit the `.txt` file directly and re-upload each time). When using the GUI’s editor (Fig. 4), after varying parameters always click on **Save device parameters** before running the simulation again. Modify the key absorber parameters, and check also the contact settings in the simulation setup file. The aim is to bring the device power conversion efficiency above 25% (i.e., MPP above 250 W m^{-2}), as close as possible to the target JV curve in Fig. 5.

Parameter	Symbol	Starting value	Unit
Thickness	L	100	nm
Relative permittivity	ϵ_r	24	—
Conduction-band edge	E_c	3.90	eV
Valence-band edge	E_v	5.33	eV
Effective DOS (CB and VB)	N_c	2.2×10^{24}	m^{-3}
Electron / hole mobility	μ_n, μ_p	1×10^{-4}	$\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$
Bulk trap density	$N_{t,\text{bulk}}$	2×10^{22}	m^{-3}
Interface trap density (to HTL)	$N_{t,\text{int}}$	5×10^{15}	m^{-2}
Band-to-band recombination	k_{direct}	3×10^{-17}	$\text{m}^3 \text{ s}^{-1}$
Optical absorption file	<code>nkLayer</code>	<code>nk_absorber_Eg1.43eV.txt</code>	

Table 1: Starting values of the absorber parameters in `bad.start/ABS_parameters.txt`. The bandgap and band-edge alignments with the transport layers are already set; the other parameters are the ones that may need tuning.

Select a file to edit

ABS_parameters.txt

▼ General

L	100E-9	m, device length/thickness
eps_r	24	relative dielectric constant
E_c	3.90	eV, conduction band edge
E_v	5.33	eV, valence band edge
N_c	2.2E24	m ⁻³ , DOS of conduction and valence bands
N_D	0	m ⁻³ , ionised n-doping
N_A	0	m ⁻³ , ionised p-doping

▼ Mobilities

mu_n	1E-4	m ² /Vs, zero field mobility
mu_p	1E-4	m ² /Vs, zero field mobility
mobnDep	0	0 : const. mob, 1 : field-dependent
mobpDep	0	0 : const. mob, 1 : field-dependent
gamma_n	0	(m/V) ^{0.5} , field dependence of mob, Poole-Frenkel form
gamma_p	0	(m/V) ^{0.5} , field dependence of mob, Poole-Frenkel form

Figure 4: Text editor of parameter and simulation setup files available in the GUI. To edit the simulation and layer parameters, change values in the text boxes and always click **Save device parameters** afterwards.

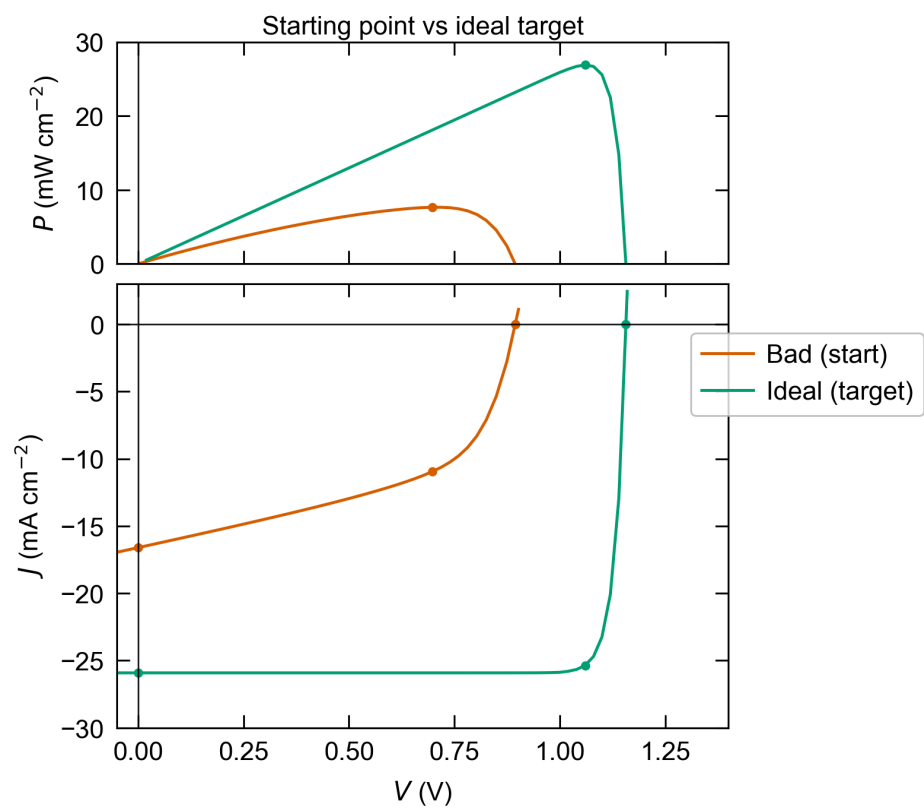


Figure 5: Starting point (orange) vs. ideal target (green).